

# AWS EC2 & S3 Practical - M Tabish

**Objective:** To deploy a simple web server using **AWS EC2 + Ubuntu + Nginx** and upload an image to **AWS S3** with a public URL.

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## EC2 Web Server

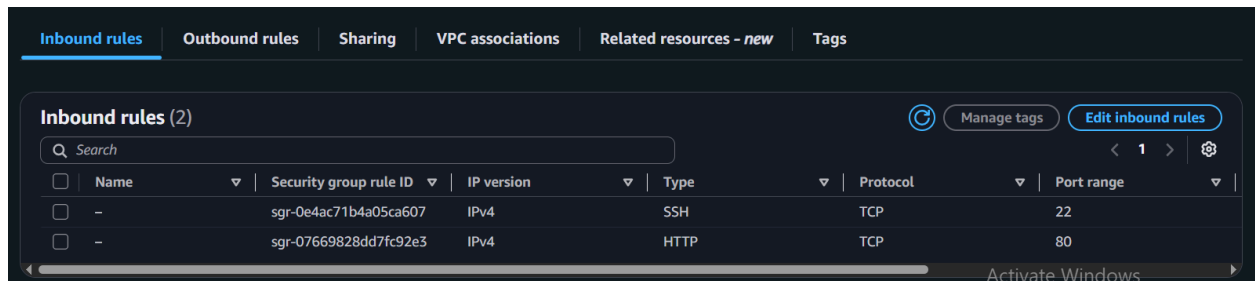
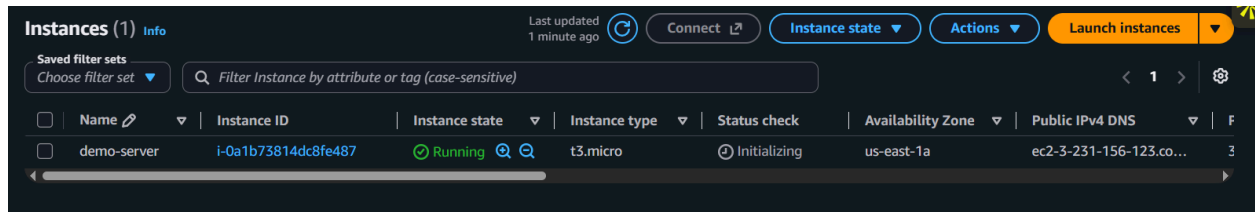
### 1: Create EC2 Instance

I selected **Launch Instance** and configured an Ubuntu server.

#### Configuration:

- OS: Ubuntu
- Instance Type: t3.micro
- Key Pair: **demo-server-key**
- Security Group: SSH (22) and HTTP (80)

**Purpose:** To create an Ubuntu virtual server for hosting a webpage.



## 2: Fix SSH Key Permissions

I received a private-key permission error, so I changed the permissions of the `.pem` file. For this I use the below mentioned commands.

- `icacls "C:\Users\Dell\Downloads\demo-server-key.pem" /inheritance:r`
- `icacls "C:\Users\Dell\Downloads\demo-server-key.pem" /grant:r "%USERNAME%:R"`

Then I connected again using SSH.

**Purpose:** To allow SSH to securely use the private key.

## 3: Connect to EC2 using SSH

I downloaded the `.pem` key and connected to the EC2 instance from Windows CMD.

```
ssh -i "C:\Users\Dell\Downloads\demo-server-key.pem" ubuntu@http://3.231.156.123/
```

**Purpose:** To remotely access the Ubuntu server.

```
ubuntu@ip-172-31-9-203: ~  
  
C:\Users\Dell\Downloads>ssh -i "demo-server-key.pem" ubuntu@3.231.156.123  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Fri Sep 25 10:35:56 UTC 2026  
  
System load:  0.08      Temperature:   -273.1 C  
Usage of /:   30.5% of 6.61GB  Processes:    117  
Memory usage: 25%      Users logged in: 0  
Swap usage:   0%        IPv4 address for ens5: 172.31.9.203  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
ubuntu@ip-172-31-9-203:~$  
ubuntu@ip-172-31-9-203:~$
```

## 5: Update Ubuntu

sudo apt update

**Purpose:** To update the Ubuntu package information before installing software.

## 6: Install Nginx

sudo apt install nginx -y

**Purpose:** To install Nginx as the web server.

## 7: Check Nginx

sudo systemctl status nginx

**Purpose:** To confirm that Nginx is running successfully.

```
no VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-9-203:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-09-25 10:40:19 UTC; 33s ago
     Invocation: 748f9b0301ca4804be2811535e58a4e7
       Docs: man:nginx(8)
    Process: 1859 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 1861 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 1890 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3M (peak: 7M)
        CPU: 43ms
    CGroup: /system.slice/nginx.service
            └─1890 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
               └─1893 "nginx: worker process"
                  └─1894 "nginx: worker process"

Sep 25 10:40:19 ip-172-31-9-203 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Sep 25 10:40:19 ip-172-31-9-203 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-9-203:~$
```

## 8: Now I go to Website Directory

cd /var/www/html

Then:

ls (to list all directories)

**Purpose:** To access the directory where Nginx stores website files.

## 9: Edit Custom index.html

I edited the webpage using:

sudo nano /var/www/html/index.html then I added my custom HTML code.

```
ubuntu@ip-172-31-9-203:~$ sudo nano /var/www/html/index.html
ubuntu@ip-172-31-9-203:~$ ^C
ubuntu@ip-172-31-9-203:~$ curl http://localhost
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">

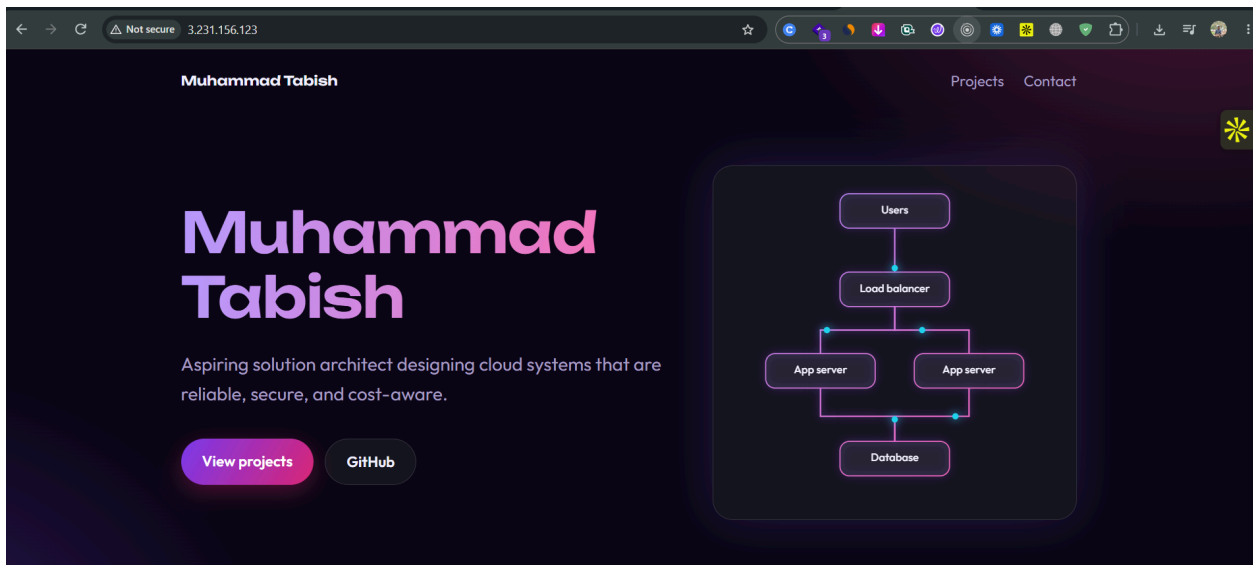
  <!-- EDIT: page title and description -->
  <title>Muhammad Tabish | Aspiring Solution Architect</title>
```

## 10: Access Website

I opened the EC2 public IP in my browser:

<http://3.231.156.123/>

The custom `index.html` page displayed successfully



# Part 2 — Amazon S3

## 1: Create S3 Bucket

I opened **AWS Console** → **S3** and created a new bucket.

Bucket Name: demo-image1

**Purpose:** To create cloud storage for my image.

## 2: Upload Image

I opened the bucket and selected:

Upload → Add Files → Select Image → Upload

**Purpose:** To store the image directly in S3.

## 3: Configure Public Access

I configured the required S3 public-access settings and permissions for the image.

For the bucket policy, the public read permission was configured as:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadImages",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::YOUR-BUCKET-NAME/*"
    }
  ]
}
```

**Purpose:** To allow the uploaded image to be accessed publicly.

## 4: Get S3 Public URL

I opened the uploaded image and copied its **Object URL** in my browser.

**Result:** The uploaded image was displayed successfully.